A SPECTRAL MAP IS CONSTRUCTIBLY CONTINUOUS, HAS QUASI-COMPACT FIBRES AND CONSTRUCTIBLY CLOSED IMAGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that the constructible topology of a spectral space is Hausdorff and quasi-compact. Invoked at the start of the counted proof (source lines 4599-4601). Puts both spaces in the setting of quasi-compact Hausdorff spaces.
- The statement that a quasi-compact subset of a Hausdorff space is closed. Invoked inside the counted proof (source lines 4605-4606). Gives closedness of the constructible image.
- The statement that a closed subset of a quasi-compact space is quasi-compact. Invoked inside the counted proof (source lines 4613-4614). Makes the constructible fibres quasi-compact.
- The statement that quasi-compactness transfers across a surjective continuous map. Invoked at the end of the counted proof (source line 4616). Gives quasi-compactness of the fibres in the given topology.

Source reference. Original lemma lemma-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1879–1879. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-constructible-hausdorff-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 4522–4522. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-hausdorff-quasi-compact, which is an existing labeled result of this same chapter

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Source reference. Original lemma lemma-image-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1964–1964. Navigation pointer only: the counted text refers to the Stacks item lemma-image-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-quasi-compact-in-Hausdorff: sources/stacks-topology/pinned-original-source.tex, source lines 1894–1894. Navigation pointer only: the counted text refers to the Stacks item lemma-quasi-compact-in-Hausdorff, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : X \to Y$ be a spectral map of spectral spaces. Then*

(1) *f is continuous in the constructible topology,*
(2) *the fibres of f are quasi-compact, and*
(3) *the image is closed in the constructible topology.*

Proof. Let X' and Y' denote X and Y endowed with the constructible topology which are quasi-compact Hausdorff spaces by Lemma lemma-constructible-hausdorff-quasi-compact. Part (1) says $X' \to Y'$ is continuous and follows immediately from the definitions. Part (3) follows as $f(X')$ is a quasi-compact subset of the Hausdorff space Y', see Lemma lemma-quasi-compact-in-Hausdorff. We have a commutative diagram

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \end{array}$$

of continuous maps of topological spaces. Since Y' is Hausdorff we see that the fibres X'_y are closed in X'. As X' is quasi-compact we see that X'_y is quasi-compact (Lemma lemma-closed-in-quasi-compact). As $X'_y \to X_y$ is a surjective continuous map we conclude that X_y is quasi-compact (Lemma lemma-image-quasi-compact). □